

Trustworthiness in Qualitative Research

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Introduction

The only statistics you can trust are those you falsified yourself.

—Attributed to Winston Churchill but, ironically, may be falsified.

He uses statistics as a drunken man uses lamp posts—for support rather than for illumination.

—Andrew Lang

To be relevant, all research must be trustworthy. Although quips about statistics are often cynical—as numbers can be manipulated by those with an agenda—at least quantitative research has the facade of being intrinsically trustworthy. This is because the quantitative research process is more systematic and apparent, and finding relationships between numbers is assumed to be more rigorous than deriving meaning from words and actions. Qualitative research, for reasons that will be laid out in this article, is considered by many in the scientific community to be less credible. This perception, whether real or imagined, makes trustworthiness a compelling goal. In this article, I will explore what it means for a qualitative study to be trustworthy, how trustworthiness can be attained by researchers, and how lactation professionals can assess the studies they read for trustworthiness.

Qualitative Research Defined

In the broadest of terms, research is defined by its methods—those relying on quantifiable measurement and those relying on description without numerical referents. Research is a process to discover new knowledge. The aim of qualitative research is to understand meaning, and the data to be analyzed are words and observations. For an anthropologist like myself, qualitative research is not some frivolous pursuit, nor is it a peripheral enterprise that complements the real work of knowledge production: qualitative research IS the principal means of knowledge production. I think that it is productive to review some of the epistemological bases of qualitative research, and how this has been debated.

First, it is helpful to understand the difference between a research method and epistemology—or assumptions about

the nature of knowledge production. Although this may appear obvious, confusing the two is common and it leads to misunderstandings. Method is how one does research. Examples of qualitative research methods are ethnography, grounded theory, case study analysis, and content analysis. These are all techniques that researchers use to conduct their inquiry. In ethnography—the hallmark research method of anthropology—the researcher spends extensive time in the field conducting participant observation among the research participants and producing field notes, which are the data that will later be analyzed. Hence, ethnography is the method. There can be a correlation between epistemology and the method selected, but this is not necessarily the case.

Science itself is a method. It is a method with certain epistemological and ontological assumptions underlying it, but it is a method, nonetheless. The assumptions made by those who employ the scientific method are twofold: (1) The world is knowable via our five senses; and (2) the truth of this knowledge is confirmed through verification. The scientific method is implemented through a systematic series of steps that can be replicated by others. When employing the scientific method, the investigator formulates a hypothesis about the relationship between two or more variables. The research study is a way to test hypotheses. The aim of a study should not be to prove a hypothesis. Rather, skepticism is employed, and the goal is to try to disprove a hypothesis. Karl Popper (2014) made this abundantly clear in his famous parable about black swans—one does not demonstrate the truth claim, “all swans are white” by looking for white swans. This only leads to the logical fallacy of confirmation bias, whereby it is a certainty that the researcher will find exactly what he or she was looking for in the first place (McSweeney, 2021).

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Rather, the best way to demonstrate that all swans are white is to look for black swans. If the hypothesis is not disproven, the hypothesis stands. The hypothesis is accepted as a truth claim *only as long as it is not disproven*. Hence, knowledge is produced by scientists who implement skepticism to test hypotheses and try to disprove them. The more a truth claim stands up to attempts to disprove it, the stronger the truth claim becomes. Science is an ongoing process of seeking truth and employing an attitude of skepticism, as well as an organized series of steps that are independent of any one investigator. Truth claims are confirmed through verification—they are verified by others making the same observations and coming to the same conclusions. That is, they are replicable.

The set of epistemological assumptions of a researcher is something different from the methods that the researcher employs. What makes qualitative research different from quantitative research is that, because qualitative researchers are interested in finding meaning—and meaning allegedly involves more subjectivity than measurement—qualitative researchers are more likely than quantitative researchers to be dismissive of the scientific method and its quest for objective truth. Also, the scientific method is mostly deductive. One sets out to test a hypothesis. Qualitative researchers can use deductive reasoning and test hypotheses in their work. However, more often, qualitative researchers prefer an inductive approach—one which starts with an open question and allows for new discovery that is not limited by formulaic hypothesis testing. Yet another issue is that in qualitative research, the researcher is often the principal research instrument (Dodgson, 2019). To return to the example of ethnography, the ethnographer goes out into the field, makes observations, records them, and produces field notes, which are later analyzed. Note, however, that this gives the ethnographer a lot of responsibility. The consumers of ethnography must have some level of trust that the ethnographer is making accurate observations, that these observations are not biased, and that the ethnographer is being intellectually honest about what is observed. Ethnographic research is not truly replicable because no two researchers are alike.

The epistemological assumptions of knowledge production in anthropology, as well as in the other social sciences, have been passionately debated. For example, some anthropologists see anthropology as a science. Those scholars want to employ the scientific method and use the tools of science to produce knowledge. Other anthropologists question the use of science to produce knowledge. Although I am simplifying the many approaches on this side of the debate by grouping them together and highlighting their similarities, all share a vision of anthropology as leaning more into the humanities than the sciences. Constructivists, interpretivists, and postmodernists all question objectivity, and whether the world is knowable; some go as far to question the existence of truth itself. They do not view the world as objectively knowable: they see different subjective truths, all equally

valid. For some of those scholars, it is an imbalance of power that privileges one version of truth over others. In bringing forth truths that are not considered conventional, research becomes a political act. It has an agenda. This is the antithesis of the scientific method, which is, by nature, objective, dispassionate, and apolitical.

Since qualitative methods are just that—a methodology, not an epistemological assumption—it is possible to conduct qualitative inquiry that employs scientific principles and methods, and it is also possible to do qualitative inquiry that rejects principles and methods. In either case, there must be some way for consumers of research to assess its quality. It is easier to assess qualitative research that adopts scientific principles because the work is more systematic, employing the conceptual toolkit of science, and can be evaluated just like quantitative studies. How representative was the sample? What was the sampling technique used? How were data analyzed? In qualitative inquiry that rejects the scientific method, assessment is much more difficult. In much of that work, the only way for consumers to evaluate the research is to gauge the aesthetics of the study, or to accept the study's conclusions based upon the authority of the researcher. This is a place where trustworthiness of qualitative research may come into question.

Trustworthiness

There is a history to the use of trustworthiness as pertains to the assessment of qualitative methods. The seminal work of Lincoln and Guba (1985) made a case for assessing qualitative research not with scientific measures, for example, reliability and validity, but, rather, upon trustworthiness. For them, trustworthiness should be evaluated by looking at credibility, transferability, dependability, and confirmability (Lincoln & Guba, 1985). Their writing generated a voluminous literature debating how qualitative studies should be evaluated (e.g., Collingridge & Gantt, 2019; Connelly, 2016; Noble & Smith, 2015; Stahl & King, 2020) that some argue, has created more confusion than clarification (Morse et al., 2002).

More recently, there have been calls to replace the term trustworthiness with rigor (Cypress, 2017), and/or continue to use the terms reliability and validity (Collingridge & Gantt, 2019; Cypress, 2017; FitzPatrick, 2019; Lueng, 2015; Morse et al., 2002). The problem is that, regardless of the terminology used, researchers rejecting the epistemological principles of scientific knowledge production do not lend themselves well to empirical methods of evaluation, especially using the conceptual toolkit of the scientific method. This is where some have made a case for evaluating qualitative studies based on aesthetics (Rolfe, 2006), something patently rejected by others as unscientific (Porter, 2007), and with good reason.

I contend that trustworthiness is crucial to assessing qualitative research and that its most important aspect is

transparency. Why is this, and what is there to be transparent about? When quantitative researchers conduct their studies, it is assumed that they are adopting the ontological and epistemological principles of science. Their conceptual toolkit implies it. The systematic methods that they use also add credibility, because the process is more important, and easier to evaluate, than the personal characteristics of the researcher. In qualitative research, however, the researcher is the principal research instrument (Dodgson, 2019). If consumers do not find the researcher trustworthy, the entire enterprise is in question. Also, as previously mentioned, some qualitative researchers reject the conceptual framework of science, because it is too limiting, too formulaic, at odds with the creative and flexible nature of inductive inquiry, or even for political reasons. Regardless of the approach of the researcher, the key to trustworthiness of a qualitative study is transparency, and by transparency, I mean that not only should the research techniques be precisely spelled out, but also that the epistemological and theoretical bases of the work must be made explicitly apparent.

Making Qualitative Research Trustworthy

Show Me the Theory

Most social science research, whether quantitative or qualitative, whether deductive or inductive, has a theoretical framework; and, to be transparent, that framework should be spelled out for the consumers of the research. Even in an inductive study, for example, grounded theory inquiry in which the investigator is looking to discover something new rather than testing hypotheses, there is an overarching perspective that guides the questions that the researcher is asking (Malterud, 2001). This is the theoretical framework or orientation. Denying the existence of this theoretical orientation, or downplaying it, is a source of potential distrust, and this is a place where scholars must be more self-aware and responsible about how they present their work.

A major problem with some qualitative inquiry is the way fact and theory are conflated. When theoretical frameworks are presented as truths, and theoretically derived concepts as unquestionable facts, it demonstrates a lack of transparency that threatens the credibility of qualitative research (Malterud, 2001; Redstone, 2019; Rose & Johnson, 2020). Furthermore, some theoretical perspectives are political in nature. The research process itself is seen as a way to implement social change, however that is understood and encouraged by that theory. Take, for example, the theory of radical feminism (see Crow, 2000). Radical feminism holds that the oppression of women is structural, as it is built into society's dominant institutions and systems. For the proponent of radical feminism, the aim of social science research is to uncover the implicit ways that women are oppressed, in order to change the system—a political aim. When scholarly

advocates of radical feminist theory do research, they are much more likely to do qualitative than quantitative work, due to the nature of the phenomena that they study—they are interested in meaning. Within the social sciences, there are countless theories that pertain to all aspects of the human condition—race, class, gender, family, religion, and so forth. Just within gender studies, there are numerous theories of feminism, each with its own set of assumptions and conceptual frameworks. This is a good thing, because it means that scholars have different ideas of how society works, and they develop theories to try to better understand that. It gets us closer to truth. Nor is change-based research necessarily a problem. Research that solves real world problems is research that makes a difference. The entire point of translational research, for instance, is to improve clinical outcomes. The problem arises when researchers with a particular theoretical perspective do not disclose to consumers that they are using one theory, *among many*, to formulate a research question, analyze data, and come to conclusions. This lack of transparency is highly problematic because it makes the work untrustworthy.

Show Me the Methods

It is also crucial for qualitative researchers to be transparent about their methods. Unlike quantitative research, the process of qualitative research is highly variable, and often involves a sole researcher who does the data collection and analysis. It is also more difficult to verify the findings of the work. There are ways for researchers to make their methods more trustworthy and I discuss several of the most important ones here. At the very minimum, laying out the exact procedures of a qualitative study—and justifying why those procedures were selected—is fundamental to assessing the soundness of the research (Rose & Johnson, 2020). Hence, the suggestions listed here should be explicitly stated in the write up of research.

Reflexivity. In her article about this topic, the editor in chief of *JHL* shows how reflexivity is key to assessing trustworthiness in qualitative research (Dodgson, 2019). Reflexivity means that researchers should reflect and share with readers how their social positions vis a vis research participants may interfere with the researcher's ability to maintain objectivity. The effects of the researcher on the research participants and research participants on researchers are both potential areas of bias and misunderstanding. As an anthropologist, it is by design that my research participants are substantially different from myself, and as I conduct my research I reflect not only on how our cultural differences might lead to my misunderstanding them, but also how my social positions are perceived by them (my class, gender, religion, citizenship status etc.) and might lead them to change what they say or do in my presence. Researchers are only human, and as professional as they might be, their own perceptions, understandings, and

biases make pure objectivity impossible. Sometimes, being an outsider is a positive—insiders may be likely to share information simply because the researcher is not native and will not feed that information back into the group. That said, objectivity should be the goal—we ought to be doing the best we can. Reflexivity is a form of intellectual honesty. Therefore, during all stages of research the researcher should self-assess and consciously try to minimize areas where biases might cloud objectivity. In the write up, the researcher can explain this process to the consumers of the research, making the study's trustworthiness more apparent.

Triangulation. Triangulation is cross-checking research to make it more credible (Patton, 1999; Stahl & King, 2020). There are various ways to do this. If working with a team, team members can cross check each other's work; this is investigator triangulation (Patton, 1999). For example, one researcher can transcribe another's interviews to ensure consistency in interviewing techniques, or transcriptions can be transcribed and cleaned by more than one member of the team. Another way of ensuring accuracy is to have research participants read and provide feedback about the analysis and the results, to ensure that the findings of the researchers are accurate from a native perspective. This form of triangulation is known as member-checking (Stahl & King, 2020). Methodological triangulation is another procedure for ensuring accuracy, this is when a researcher uses multiple methods (usually three) to ensure validity (Patton, 1999). If studying attitudes about breastfeeding of immigrant women, for example, interviews might be the main source of data, but the researcher would also conduct focus groups and ethnography to validate the information in the interviews. Finally, theory triangulation applies different theoretical frameworks to the data, or, in evaluation research, views data from the lenses of different stakeholders (Patton, 1999). These various forms of triangulation can all contribute to the trustworthiness of qualitative research.

Raw Data Availability and Data Tables. One of my anthropology professors wisely taught me that, in qualitative research, nothing is considered data until it is written down. That is, when anthropologists conduct ethnographic fieldwork, there are no data until fieldnotes are recorded. Indeed, one of the challenges of qualitative research is the sheer amount of data it produces. Interviews and focus groups yield transcriptions, ethnography, and other forms of observation yield field notes. To analyze their data, qualitative researchers must spend time poring through a substantial volume of information.

There are two ways that this data can be used to increase trustworthiness. One is to make the raw data available to consumers of research. Of course, the data might have to be deidentified for ethical reasons but making data—for example, interview transcripts—available to consumers of research can allow them to assess for themselves the accuracy of the analysis and findings. Another way is to present as much of the data and its analysis in table form in the manuscript

(Cloutier & Ravasi, 2021). This visual representation provides readers with a better understanding of how researchers came to their findings. In short, it makes the research process more transparent, and therefore trustworthy.

Conclusion: Why do Lactation Professionals Consume Research?

Lactation clinicians consume research to improve their practice. This depends on getting trustworthy information—which means reliable and valid information. In the parlance of current health literature, the most worthwhile and valuable research is evidence-based (Robinson et al., 2021). This means that research builds upon prior research. Each study should be designed with previous research in mind, and new findings should be understood in the context of preexisting literature. Evidence-based information is considered trustworthy information, but are all studies producing evidence-based findings? If a researcher grounds a study in current literature, which most qualitative studies do, but rejects the epistemological basis of scientific knowledge production, how can those findings be considered evidence-based? In some cases, what qualitative scholars produce should not even be considered research. An example of that is the post-modernist technique of autoethnography, in which a scholar puts him or herself at the center of inquiry—the data, the analysis, and the conclusions are all generated from the writer (Ellis & Bochner, 2000). There is no objectivity possible in this work. There is no way to validate or confirm it. It is accomplished in a way that rejects the scientific method. I know that this may be a controversial position to hold, but I question whether this can even be called research.

That said, is there something that can be learned about lactation from sources besides research? As an anthropologist, I believe it is undeniable that we have much to learn about our patients and how we care for them from literature, art, poetry, music, religion, and oral tradition. These are all valuable sources of knowledge. However, they are not research. When scholars conduct qualitative inquiry that rejects scientific principles of objectivity and do not adhere to some standard of scientific process and rigor, that work, while still valuable in other ways, is not trustworthy as research. When qualitative researchers reject science, they can only be assessed by aesthetics or by the qualifications of the researcher (Porter, 2007). These are not trustworthy either.

The same can be said for research with a social agenda—but only when that agenda is not explicitly made. Lactation professionals who take an advocacy role can benefit from theoretical orientations with an action-based, political agenda; but, to be trustworthy, that agenda must be transparent. Even so, the product of this research cannot be considered evidence-based, because if there is a political agenda underlying it, it violates the scientific principle of objectivity. True science favors no one; it is dispassionate. The results can make people very unhappy when they challenge their own understandings about the way things are. This is a

Table 1. Assessing Trustworthiness.

Questions to Answer
Is the author adopting or rejecting the scientific method?
Which theoretical framework is guiding the research?
Which methods are the researcher using to collect data and analyze data?
Does the author have a social or political agenda, and is that stated?
How is the author ensuring maximum objectivity and resisting bias?
How is the researcher maintaining rigor in the process of the study?

broader issue within the social sciences at large—but it comes out very clearly when examining the issue of trustworthiness in qualitative research.

Lactation professionals are both researchers and consumers of research. When assessing the work of others, or when designing a study, consideration of the questions in Table 1 are a fine measure of trustworthiness. These issues are pertinent to transparency. The take-home point of this article is that for qualitative studies to be considered trustworthy, they must be transparent in both theory and method.

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